

Large Language Models in Mental Health: An Investigation of Prompt-Based Approaches, Fine-Tuning and Domain Adaptation, and Graph Retrieval-Augmented Generation

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Abstract. Large language models have shown growing potential in mental health services, including psychological support and knowledge-based therapeutic assistance. This paper reviews three main technical approaches in this area: prompt engineering, domain-specific fine-tuning, and Graph Retrieval-Augmented Generation. Prompting is flexible and relatively low-cost, which makes it useful for early exploration. Fine-tuning can improve the model's relevance to mental health tasks by using domain-specific data, while Graph-RAG helps strengthen contextual understanding and makes responses better grounded in external knowledge. At the same time, several important problems remain. Current LLM-based systems still face limits in interpretability and traceability, and issues such as hallucination and safety risks have not been fully resolved. Another major challenge is the lack of strong clinical validation, since many existing studies focus more on technical performance than on real-world treatment outcomes. To address these problems, future research may benefit from adding human supervision, exploring agent-based systems, and carrying out more rigorous clinical and real-world evaluation. Overall, although LLMs show clear promise in mental health, their development into safe, reliable, and clinically useful tools will still require continued technical improvement, professional involvement, and appropriate regulation.

Keywords: Large language models, deep learning, mental health

1. Introduction

In recent years, the application of Large Language Models in mental health has become an emerging and rapidly growing research area. Mental health, as the foundation of a person's overall health, plays a crucial role in one's growth and development throughout life. However, due to the widespread neglect and prejudice towards mental health treatment among both the nation and society, the mental health field is currently facing a shortage of medical staff, and differences in economic and social development across countries and regions have led to unequal attention to mental health. The imbalance in the development of psychological therapy is becoming increasingly severe. Especially in impoverished and underdeveloped areas, they often cannot afford the high costs of treatment and the time required. According to the WHO Mental Health Atlas 2024, the global median government expenditure on mental health was US\$ 2.69 per capita, but this varied

sharply across income groups, from US\$ 0.04 in low-income countries and US\$ 0.34 in lower-middle-income countries to US\$ 65.89 in high-income countries [1]. Therefore, the traditional psychological therapy system is unable to meet the increasingly growing psychological health needs of people. Open-source or low-cost large language models are expected to alleviate the problems.

In the early research and applications of Large Language Models in mental health, researchers mainly interacted with large language models through dialogue and different prompting strategies. For example, Souza et al. developed a prompt engineering framework for psychological consultation and showed that layered, empathy-driven, and scenario-based prompts could produce more relevant, compassionate, and contextually accurate responses. Their findings suggest that early prompt-based Large Language Model applications in mental health were promising, yet still exploratory in terms of reliability and practical effectiveness [2]. Although prompt-based approaches have the advantages of high efficiency, convenience, and low cost, their professionalism and empathic ability cannot always be guaranteed. Therefore, the research focus gradually shifted toward fine-tuning and domain adaptation approaches, which aim to improve the professional relevance and empathic ability of Large Language Models through fine-tuning. Chen et al. reported that fine-tuning LLMs on a large-scale empathy dialogue dataset significantly improved empathy- and listening-related abilities [3].

Through the introduction of domain-specific datasets and adaptation strategies, fine-tuning and domain adaptation approaches have guided the application of Large Language Models in mental health toward a more professional and safer direction. Although these methods are supported by dataset-based evidence, problems such as hallucinations and the lack of traceability in generated dialogue remain. On this basis, research has further evolved toward knowledge-grounded methods such as Graph Retrieval-Augmented Generation. For example, Luo et al. proposed a graph-enhanced retrieval framework to improve the quality of knowledge retrieval and reasoning, showing that recent studies have increasingly focused on strengthening factual grounding through structured knowledge representations [4]. Open-source or low-cost large language models have the potential to alleviate the current challenges facing mental health services, especially staff shortages, high costs, and unequal service accessibility across different regions. Large Language Models have the potential to transform traditional mental health care from a resource-constrained model reliant on limited expert availability into a scalable form of algorithmic support.

Currently, Large Language Model technology has made contributions in areas such as psychological counselling and emotional support chat, screening and identification of psychological problems, clinical training, supervision, and research assistance for psychological therapists. However, in practical applications, the combination of Large Language Model and mental health also faces challenges related to security, ethics, and actual effectiveness. This article discusses three major Large Language Model-based approaches in mental health: Prompt-based approaches, Fine-tuning and domain adaptation approaches, and Graph Retrieval-Augmented generation. It will also present the application effects of these approaches and challenges. Furthermore, it will explore the future development of Large Language Model applications in mental health. To provide a clearer overview of the development pathways discussed in this review, Figure 1 presents the technical roadmap of LLM-based approaches in mental health support. This article contributes to clarifying the strengths and limitations of different technical strategies.

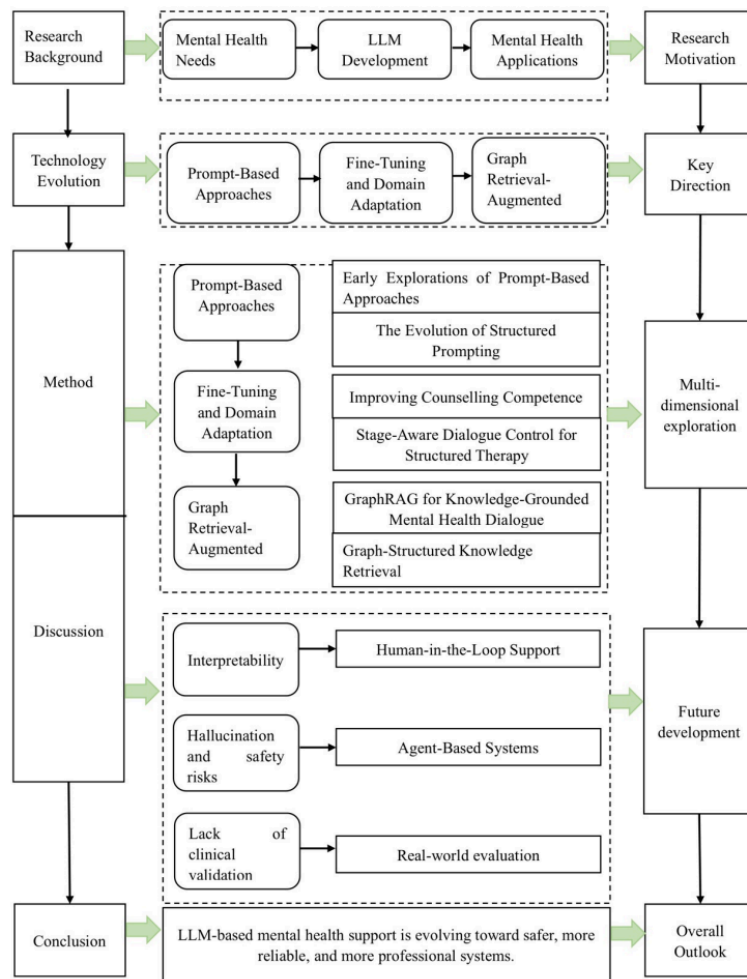


Figure 1. Technical roadmap of large language models in mental health support (picture credit: original)

2. The application of Large Language Models in mental health

2.1. Prompt-based approaches

Since 2022, with the release of large language models such as ChatGPT, researchers have begun to explore the application of LLMs in mental health services. Among the earliest research focuses was the use of zero-shot and few-shot prompting to provide mental health support. In the initial stage, researchers mainly relied on simple natural language instructions to test the feasibility of such applications. As research progressed, these simple prompting strategies gradually evolved into more systematic service structures aimed at forming professional psychological intervention workflows. However, although these methods are relatively low-cost to deploy and highly efficient, they still face important challenges, including unstable dialogue generated by prompts, the need for clearer boundaries and safety control over conversational content, as well as broader ethical concerns.

2.1.1. From general conversation to simulated empathy: early explorations of prompt-based approaches

In early Large Language Model research, a major focus was on how to design prompts so that models could meet the conversational requirements of chatbots in specific task settings while balancing flexibility and user experience. Kumar et al. framed this as a practical HCI challenge. To systematically explore prompt design, they decomposed prompts into three independently controllable dimensions: identity, intent, and behaviour, and compared 18 prompt combinations through a randomised factorial experiment with 945 valid user responses. The study tested "Coach" and "Friend" identities and found that different intents, including non-directive support, CBT, and problem-solving therapy, produced observable effects on conversational dynamics. Qualitative analysis showed that GPT-3 displayed several empathy-like behaviours under prompt guidance, such as restating users' concerns, making them feel heard, and responding positively. However, these behaviours were not explicitly specified by the prompts but emerged naturally [5]. This suggests that prompting could shape conversational style, but its empathic effects were emergent rather than fully controlled.

2.1.2. From emotional support to clinical tasks: the evolution of structured prompting

Filenko et al. applied prompt engineering to ChatGPT-4 and implemented Problem-Solving Therapy (PST) in specific clinical stages, focusing on two steps: symptom identification and assessment, and goal setting, to assist clinical treatment. The study constructed seven model configurations by progressively adding four prompt techniques, namely Baseline, Zero-Shot, Few-Shot, and CoT, and these configurations were blindly evaluated by seven clinicians. The results showed that models incorporating Few-Shot prompting outperformed the human rule-based system in both symptom assessment and goal setting. However, Filenko et al. found that while Chain-of-Thought prompting enhanced empathy, particularly in exploration, it weakened task structure by reducing the quality of symptom identification and goal setting. This finding reveals the inherent tension between structure and empathy and demonstrates that LLMs have high accuracy in symptom identification and can produce assessment results consistent with PST guidelines [6].

2.2. Fine-tuning and domain adaptation approaches

To overcome the inconsistency of prompting mentioned in Section 2.1, researchers gradually shifted toward a fine-tuning approach based on vertical-domain data. The essence of prompt engineering lies in activating the model's general pretrained capabilities; however, when addressing the highly specialised and logically rigorous demands of psychological therapy, it often exhibits limitations in terms of clinical adaptability.

Domain fine-tuning, through targeted architectural design and parameter adjustment, makes use of high-quality datasets from specific industries for training, enabling the model to achieve deep alignment with domain knowledge. This approach not only equips the model with a more professional knowledge base but also enables it to adhere more strictly to specific therapeutic protocols. The shift from temporary instruction-based guidance to persistent parameter optimisation has become an inevitable trend in research on large language models for mental health.

2.2.1. Fine-tuning on clinical data: improving counselling competence

Zhang et al. proposed the CPsyCoun framework to address the lack of professional counselling knowledge in existing mental health datasets. The framework includes two components: a multi-turn dialogue reconstruction method based on real counselling reports, and an automatic evaluation benchmark for counselling dialogues. The study adopted a two-stage Memo2Demo framework. In the first stage, GLM-4 acted as a "psychological supervisor" and transformed structured counselling reports into counselling notes containing goals, techniques, and action plans. In the second stage, GLM-4 acted as a "counsellor" and generated multi-turn dialogues across four stages—intake, diagnosis, counselling, and closing—by strictly following the reports and notes. Internal evaluation showed that Memo2Demo improved performance over direct role-play by 53%, 53%, and 30% on three dimensions. The resulting dataset, CPsyCounD, containing 3,134 high-quality dialogues, was used to fully fine-tune InternLM2-7B-Chat, producing CPsyCounX. External evaluation showed that CPsyCounX significantly outperformed SoulChat, ChatGPT, and GLM-4 in professionalism, supporting the claim that fine-tuning on CPsyCounD enables the model to naturally acquire professional psychological counselling techniques [7].

2.2.2. Beyond fine-tuning: stage-aware dialogue control for structured therapy

Chen et al. proposed a Structured Dialogue System to address a common limitation of existing fine-tuning methods. They pointed out that many such approaches do not adequately consider the different stages of counselling, which can result in dialogues lacking direction and coherence. The paper argues that the quality of multi-turn psychological counselling does not depend only on the capability of the language model itself, but also on whether the system can manage counselling stages and track the progress of the interaction. Based on this idea, the authors did not try to improve the model itself. Instead, they built a stage-control framework to keep the dialogue moving in a more organised way. The system has five parts: a stage controller, a stage-aware instruction generator, a topic database, a pretrained LLM, and a response parser. In each dialogue turn, the instruction generator creates a prompt for the current stage using four elements: the stage instruction, earlier topic records, the user's input, and a response template. The LLM then updates the topic record, decides the dialogue state, and generates a response. Here, $c \in \{-1, 0, 1\}$ means stage rollback, staying in the same stage, or moving to the next stage. To test the system, the researchers first used GPT-4 to extract client profiles from 148 real PM+ intervention transcripts. After that, they used GLM-4 to simulate multi-turn conversations across three systems. GPT-4 was then used to evaluate the dialogues in terms of logical coherence, professionalism, empathy, and realism. The results showed that the Structured Dialogue System did better than CPsyCounX in coherence, suggesting that the stage-based design helped make the dialogue flow more smoothly [8].

2.3. Graph Retrieval-Augmented Generation

Although fine-tuning and domain adaptation have improved LLM performance in some mental health tasks, these methods still depend largely on the model's internal parameters. Because of this, the source of the generated content is often hard to trace, and hallucination is still a problem. Graph Retrieval-Augmented Generation has therefore attracted more attention as another possible approach. Instead of relying only on the model's internal knowledge, it adds structured external knowledge to the generation process. In this method, knowledge is organised as a graph, so the

model can retrieve information through the relationships between different concepts. This makes the response more grounded in external knowledge and may improve both accuracy and reliability.

2.3.1. From local to global: GraphRAG for knowledge-grounded mental health dialogue

Edge et al. proposed GraphRAG to address one of the main weaknesses of conventional RAG. They argued that traditional RAG systems often struggle with global questions that require an understanding of the overall content of a large document collection. GraphRAG works through six main steps. First, source documents are divided into smaller text chunks, and a large language model extracts entities, relationships, and claims from them to build an entity knowledge graph. This graph is then organised hierarchically using the Leiden community detection algorithm, which recursively identifies smaller sub-communities until no further division can be made. Based on these graph communities, the system generates report-style summaries at different levels of the hierarchy. When a query is given, GraphRAG applies a map-reduce process to produce partial responses from different communities in parallel, rank them by helpfulness, and combine them into a final answer. According to the authors, early evaluation results showed that GraphRAG produced more comprehensive and diverse answers than a vector-based RAG baseline, and it also compared favourably with a graph-free global summarisation approach based on map-reduce over source texts [9].

2.3.2. From independent documents to networked graph structures: GRAG for graph-structured knowledge retrieval

Hu et al. proposed Graph Retrieval-Augmented Generation to address a core limitation of conventional Retrieval-Augmented Generation, namely its difficulty in capturing the topological relationships among networked documents, such as citation graphs, knowledge graphs, and social media networks. Its technical framework consists of two main stages: textual subgraph retrieval and graph-augmented generation. In the retrieval stage, Graph Retrieval-Augmented Generation first treats each node's K-hop ego-graph as a basic unit, encodes its textual attributes, and further constructs graph-level representations to support query-oriented subgraph retrieval. The system then ranks candidate subgraphs according to their similarity to the query and applies a soft-pruning mechanism to adaptively filter nodes and edges that are less relevant to the query, thereby reducing noise. In the generation stage, Graph Retrieval-Augmented Generation presents the retrieved subgraph information to the large language model from two complementary perspectives. From the textual perspective, the subgraph is converted into a hierarchical textual description through a breadth-first search-based traversal and used as a hard prompt. From the graph perspective, the topological structure is encoded by a graph neural network, and the resulting graph representations are aligned with the large language model token space as soft prompts. Hu et al. further reported that Graph Retrieval-Augmented Generation significantly outperformed Retrieval-Augmented Generation-based retrievers and large language model baselines in graph reasoning tasks, and that frozen large language models equipped with Graph Retrieval-Augmented Generation consistently surpassed fine-tuned large language models across all tasks. On the WebQSP dataset, its Hit@1 score improved by as much as 74.45% over the pure large language model baseline [10].

Table 1. Comparison of major technical approaches in LLM-based mental health support

Approach	Main Idea	Representative Features	Strengths	Limitations
Prompt-Based Approaches	Guide model responses through prompt design without changing internal parameters.	zero-shot and few-shot prompting, simulated empathy, structured prompting	Easy to deploy, low cost, flexible for early exploration	Output instability, limited controllability, weak domain specificity
Fine-Tuning and Domain Adaptation	Improve mental health dialogue performance through domain-specific training and parameter adaptation.	Fine-tuning on counselling datasets and stage-aware dialogue control.	Better professionalism, stronger empathy, improved counselling competence	Requires high-quality data, depends on training resources, limited traceability
Graph Retrieval-Augmented Generation	Introduce structured external knowledge into the generation process through graph-based retrieval	Knowledge graphs, hierarchical summarization, subgraph retrieval, and graph-augmented generation.	Better factual grounding, improved reliability, reduced hallucination	Higher system complexity, more difficult implementation, dependence on external knowledge quality

3. Discussion

Table 1 summarizes the key ideas, representative features, advantages, and limitations of the three main technical approaches described above. The discussion section, building on this comparison, will focus on the risks and challenges currently encountered in applying LLMs to mental health, and will propose feasible solutions to the identified problems, with the aim of improving the safety, reliability, and clinical utility of LLM-based mental health support systems.

3.1. Challenges

3.1.1. Interpretability

One problem shared by both prompt-based methods and fine-tuning approaches is that it is often hard to tell where the model's responses come from or why they are generated in a certain way. In real use, users may not always give very clear instructions, so the model may misunderstand what they mean. Also, because the source of the output is not always clear, the response may not connect well with the earlier conversation and may sometimes move away from the user's real question. This can reduce users' trust in the system and also makes it harder for professionals to decide whether the response is reliable or suitable in a mental health setting. Because of this, the actual value of LLMs in mental health support is still limited.

3.1.2. Hallucination and safety risks

Although research on LLMs in mental health has improved, even methods like fine-tuning and GraphRAG can still make mistakes. This is a serious problem in mental health care, where even a small error may have harmful effects. If the model gives inaccurate, inappropriate, or misleading responses, it may affect the user's emotions or treatment experience. It can also reduce people's trust in these systems and limit their value in real mental health support. As Muqtadir et al. noted, hallucination is still one of the main problems in using large language models for mental health,

because it affects both safety and reliability in counselling and therapeutic settings [11]. Because of this, reducing hallucination should be an important focus of future research.

3.1.3. Lack of clinical validation

Although many studies have shown that LLM-based systems can work well in experiments, this does not mean they are already ready for real clinical use. Up to now, a lot of research has paid more attention to technical performance, such as accuracy and retrieval quality, than to what really matters in mental health practice, like long-term treatment effects. Because of this, it is still unclear whether these systems can provide safe, stable, and professional support in real clinical settings. Jin et al. reviewed 4,859 studies and finally included 95 papers on the use of large language models in mental health. Among them, only 31 studies, or 33%, were related to clinical treatment or intervention. They also found that most studies focused on technical measures such as F1 score, accuracy, precision, and recall, instead of long-term clinical outcomes [12]. Without stronger clinical validation, research in this area may continue to stay more at the technical level than at the level of real clinical use.

3.2. Future prospects

3.2.1. Enhancing interpretability through human-in-the-loop support

To deal with the current lack of interpretability and traceability in LLM-based mental health applications, the human-in-the-loop approach may be a more realistic and safer direction. By involving professionals such as doctors or counsellors in the system, the model's responses can be reviewed and corrected when necessary. Although the model itself still cannot fully explain where every answer comes from, human supervision can make the output easier to check and reduce some of the risks in practice. Stade et al. also noted that clinical psychology is a high-risk area for AI and proposed a roadmap for using clinical large language models in psychotherapy more responsibly [13]. For this reason, the human-in-the-loop model may help improve both the reliability of the system and its practical value in real mental health settings.

3.2.2. Agent-based systems for safer mental health support

Agent-based systems may be one useful way to deal with hallucination and safety problems in current LLM-based mental health applications. Traditional LLMs usually reply one turn at a time, and it is not always easy to control the whole process. By comparison, agent-based systems divide the task into several steps, such as retrieving information, making plans, generating responses, and monitoring the interaction. This can make the whole system more structured and may also help improve the quality of the response. It can also lower the chance of giving unsupported or misleading answers. Ahalpara's "Tell Me" assistant is one example. It combines retrieval-augmented dialogue, synthetic dialogue generation, and agentic planning in the same system [14]. This shows that agent-based systems may help provide more dependable support for complex mental health needs in the future.

3.2.3. Strengthening clinical validation and real-world evaluation

Although more studies have looked at the use of large language models in mental health, there is still not enough evidence to say that these systems are ready for safe and effective clinical use.

Because of this, future research still needs stronger clinical validation and more real-world testing. These are important if researchers want to move beyond experimental studies and use these systems in actual mental health services. Real-world evaluation can also help show whether current models really fit the needs of clinical practice. At the same time, feedback from professionals can be used to improve the system and make later development more practical. Future studies should also work on clearer evaluation standards and better define the proper use of LLMs in mental health care. Ong pointed out that responsible use of LLMs in medicine still requires a sound governance framework and suitable ways to reduce risk [15]. In general, if LLMs are to move from research into clinical care, more work is still needed in clinical testing, regulation, and real-world use.

4. Conclusion

This paper discussed three main ways of using large language models in mental health: prompt-based approaches, fine-tuning and domain adaptation, and Graph Retrieval-Augmented Generation. Each method has its own role. Prompt-based methods are more flexible and usually cost less, so they are useful in the early stage of research. Fine-tuning and domain adaptation can improve the model's performance by using domain-specific data. Graph-RAG adds structured external knowledge, which can help the model produce responses that are more coherent and better grounded. At the same time, the review also shows that these approaches still share several problems. Issues such as limited interpretability, weak traceability, hallucination, and safety risks have not been fully solved. In addition, many studies still pay more attention to technical performance than to what happens in actual clinical settings. Because of this, future research should not focus only on making models stronger in a technical sense. More attention should also be given to human-centred and safe use in practice. This may include adding human supervision, exploring agent-based systems, and carrying out stronger clinical and real-world evaluation. Overall, LLMs have clear potential in mental health, but turning that potential into safe and clinically meaningful practice will still require further technical improvement, expert involvement, and ethical guidance.

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